Resistivity, temperature, power and energy

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The resistance of a conductor

In topic 1 we saw that the resistance of a wire depends on the **material**, the **length** and the **cross-section**:

> **Resistance of a conductor:** $R = \rho \cdot L / S$

Symbol	Quantity	Unit
ρ (rho)	resistivity of the material	Ω · mm² / m
L	length of the conductor	m
S	cross-section of the conductor	mm²

- if the **length** L **doubles**, the resistance **doubles**;
- if the **cross-section** S **doubles**, the resistance **halves**;
- with the same dimensions, the material with the smaller ρ has the smaller resistance: it is a **better conductor**.

For a **round** wire the cross-section is calculated from the diameter d: $S = \pi \cdot d^2 / 4$, and conversely $d = \sqrt{(4 \cdot S / \pi)}$.

Material	ρ at 20 °C (Ω · mm²/m)	Where it is used
silver	0.016	high-quality contacts
copper	**0.0175**	building wiring, windings, printed circuit board tracks
aluminium	**0.028**	overhead high-voltage lines, large distribution cables
tungsten	0.055	filaments of incandescent lamps
constantan	0.49	precision resistors, shunts
nichrome	1.1	heating elements of heaters, ovens, hair dryers

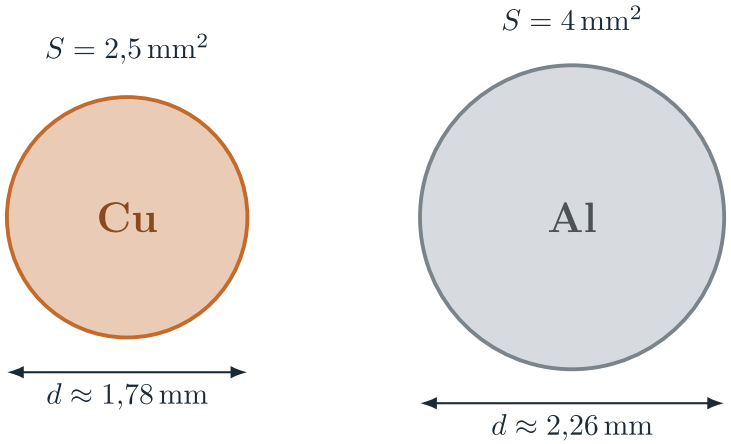
Copper or aluminium?

Copper is a better conductor than aluminium: its resistivity is smaller. To get **the same resistance** with aluminium we therefore need a **thicker** conductor. Let's see how much thicker.

Worked example: same resistance, different materials

A **copper** cable is L = 100 m long and has a cross-section S = 2.5 mm².

- **resistance of the copper cable:** $R = 0.0175 \cdot 100 / 2.5 =$ **0.7 Ω**
- **aluminium cross-section with the same resistance:** from the formula, $S = \rho \cdot L / R = 0.028 \cdot 100 / 0.7 =$ **4 mm²**
- **diameters:** copper $d = \sqrt{(4 \cdot 2.5 / 3.14)} \approx$ **1.78 mm**; aluminium $d = \sqrt{(4 \cdot 4 / 3.14)} \approx$ **2.26 mm**



What we notice:

- the aluminium **cross-section** is **1.6 times** the copper one: exactly the ratio of the resistivities, 0.028 / 0.0175 = 1.6. The ratio does not depend on the length: it holds for any cable;
- the **diameter** grows less, about 1.26 times (2.26 / 1.78): the aluminium cable is only a little thicker;
- the **weight**, instead, changes a lot. Copper weighs 8.96 g per cm³, aluminium only 2.70 g. One metre of 2.5 mm² copper cable has a volume of 2.5 cm³ and weighs about **22.4 g**; one metre of 4 mm² aluminium cable has a volume of 4 cm³ and weighs about **10.8 g**. For the same resistance, **aluminium weighs less than half**.

> **Same resistance with two materials:** $S_{Al} / S_{Cu} = \rho_{Al} / \rho_{Cu} = 1.6$

The other way round. An aluminium cable has a cross-section of 16 mm². Which copper cross-section has the same resistance? $S_{Cu} = 16 / 1.6 =$ **10 mm²**.

Why high-voltage lines are made of aluminium

Overhead high-voltage lines have very long conductors, hung from one pylon to the next. Aluminium is the better choice for three reasons:

- for the same resistance the conductor **weighs about half**, so the pylons can be lighter and further apart;

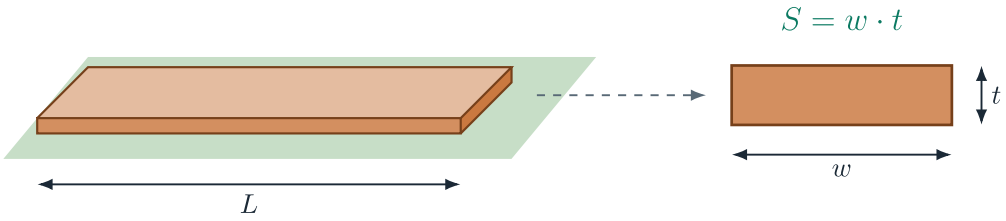
- aluminium **costs less** than copper;
- the thicker conductor cools better in the air.

Aluminium, however, is not very strong in tension: this is why line conductors are made of **aluminium and steel**, with a **core of steel wires** that carries the weight and an **outer layer of aluminium wires** that carries the current.

In **home wiring**, instead, **copper** is used: for the same resistance the cable is thinner and more flexible, it slides more easily into conduits, and the connections in the terminals are more reliable.

Printed circuit board tracks

The **tracks** of a printed circuit board are strips of copper bonded to the board. Their cross-section is not round but **rectangular**: it is w wide (*width*) and t thick (*thickness*).



> **Rectangular cross-section:** $S = w \cdot t$

The most common copper thickness is **35 µm** = 0.035 mm. The formula $R = \rho \cdot L / S$ stays the same: only the way of calculating the cross-section changes.

Worked example: resistance of a track

A copper track is w = 0.5 mm wide, t = 0.035 mm thick and L = 100 mm = 0.1 m long.

- **cross-section:** $S = 0.5 \cdot 0.035 =$ **0.0175 mm²**
- **resistance:** $R = 0.0175 \cdot 0.1 / 0.0175 =$ **0.1 Ω**

Watch the units: the length in **metres**, the cross-section in **mm²**.

Width *w*	**0.25 mm**	**0.5 mm**	**1 mm**	**2 mm**
Resistance (L = 100 mm, t = 35 µm)	0.2 Ω	0.1 Ω	0.05 Ω	0.025 Ω

A track twice as wide has half the resistance, just like a wire with twice the cross-section. If 2 A flow in the 0.5 mm track, a voltage of $0.1 \cdot 2 =$ **0.2 V** is lost across it and the track dissipates $P = 0.1 \cdot 2^2 =$ **0.4 W**, which heat it up. This is why tracks that carry **a lot of current** are drawn **wider** (or with thicker copper, for example 70 µm).

Electrical power

In topic 2 we saw the three power formulas:

> **Power:** $P = V \cdot I$; $P = R \cdot I^2$; $P = V^2 / R$

Known data	Formula to use
V and I	$P = V \cdot I$
R and I	$P = R \cdot I^2$
V and R	$P = V^2 / R$

In a conductor or a resistor all the power turns into **heat** (Joule effect). The formula $P = R \cdot I^2$ is the most useful one for **cables** and **tracks**: the current is the one the load needs, and the power lost grows with the **square** of the current. If the current doubles, the power lost in the cable becomes **four times** larger.

Worked examples: choosing the resistor power rating

Choose a resistor with a rated power at least **twice** the calculated power. Standard values: 1/4 W, 1/2 W, 1 W, 2 W, 5 W.

- **330 Ω at 12 V.** $P = V^2 / R = 12^2 / 330 = 144 / 330 \approx$ **0.44 W**. With the margin at least 0.87 W are needed: choose a **1 W** resistor.
- **100 Ω carrying 0.1 A.** $P = R \cdot I^2 = 100 \cdot 0.1^2 =$ **1 W**. With the margin 2 W are needed: choose a **2 W** resistor.
- **Copper cable from the example (0.7 Ω) carrying 10 A.** $P = 0.7 \cdot 10^2 =$ **70 W** dissipated along the cable.

Energy: from the joule to the watt-hour

Energy is power times operating time: $E = P \cdot t$. The unit of the International System is the **joule** (J): 1 J = 1 W · 1 s, that is power in watts and time in **seconds**.

The heat produced by the Joule effect in a time interval Δt is:

> **Heat (Joule effect):** $Q = P \cdot \Delta t = R \cdot I^2 \cdot \Delta t$

With time in seconds Q is in joules; with time in hours it is in watt-hours.

Why the watt-hour is used

The joule is a **very small** unit for electrical appliances. A 1200 W hair dryer switched on for 15 minutes (900 s) uses:

- in joules: $E = 1200 \cdot 900 =$ **1,080,000 J**
- in watt-hours: $E = 1200 \cdot 0.25 = 300$ Wh = **0.3 kWh**

> **Conversions:** 1 Wh = 3600 J ; 1 kWh = 1000 Wh = 3,600,000 J

The **watt-hour** and the **kilowatt-hour** are the units used throughout the world of electrical energy, that is in **distribution** and in **energy accounting**:

- the household **meter** counts the **kWh** used;
- the **electricity bill** shows consumption in **kWh** and the energy price in **€/kWh**;
- the **energy label** of appliances gives consumption in **kWh per year**;

- **batteries** (phones, e-bikes, electric cars) state the energy they hold in **Wh** or **kWh**;
- a **photovoltaic system** states the energy it produces in **kWh**;
- the consumption of a city or a whole country is measured in multiples of the watt-hour: **MWh**, **GWh**, **TWh**.

This is why, from now on, energy is always calculated in Wh and kWh, with power in watts and time in **hours**.

Resistance changes with temperature

When a metal conductor heats up, its resistance **increases**. For temperature changes that are not too large the law is **linear**:

> **Resistance and temperature:** $R(T) = R_0 \cdot [1 + \alpha \cdot (T - T_0)]$

Symbol	Meaning	Unit
$R(T)$	resistance at temperature T	Ω
R_0	resistance at the reference temperature T_0 (usually 20 °C)	Ω
α (alpha)	temperature coefficient of the material	1/°C (°C^{-1})
$\Delta T = T - T_0$	temperature change	°C

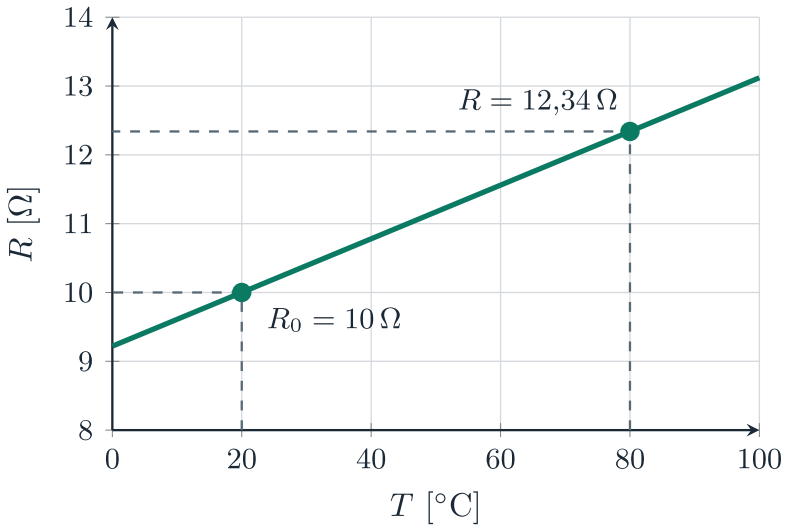
Material	α (1/°C)
copper	0.0039
aluminium	0.0040
tungsten	0.0045
constantan	≈ 0

Copper increases its resistance by about **0.4 % per degree**. **Constantan** has an almost zero α: its resistance does not change with temperature, which is why it is used in precision resistors.

Worked example: a coil heating up

A copper winding has $R_0 = 10\ \Omega$ at $T_0 = 20$ °C. During operation it reaches $T = 80$ °C.

- **temperature change:** $\Delta T = 80 - 20 =$ **60 °C**
- **hot resistance:** $R = 10 \cdot (1 + 0.0039 \cdot 60) = 10 \cdot 1.234 =$ **12.34 Ω**, that is **23 % more**

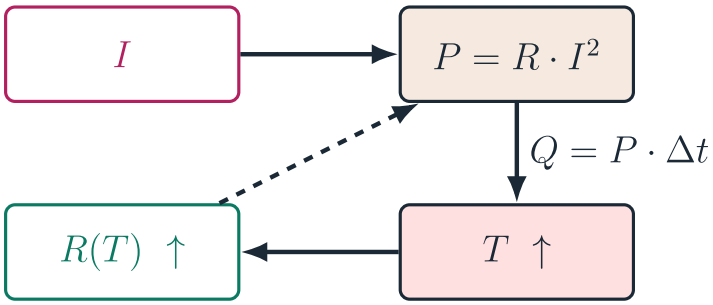


A hot cable. The 0.7 Ω copper cable at 20 °C, if it reaches 70 °C, has $R = 0.7 \cdot (1 + 0.0039 \cdot 50) \approx$ **0.84 Ω**.

Did you know? The incandescent lamp. The tungsten filament goes from 20 °C to about 2500 °C. A 60 W lamp at 230 V has, when lit, $R = 230^2 / 60 \approx 882$ Ω. With the formula we can estimate that when cold it has about $882 / (1 + 0.0045 \cdot 2480) \approx$ **73 Ω**, that is about **12 times less**. This is why at switch-on it draws, for an instant, a much higher current (about 3.2 A instead of 0.26 A), and that is exactly when it often burns out. With such large temperature jumps the linear formula gives only an **estimate**.

Joule effect and temperature: a chain reaction

The current heats the conductor (Joule effect), the hot conductor has a higher resistance, and the higher resistance changes the power dissipated. A **chain** forms:



How the power changes depends on **how the conductor is supplied**.

- **With a fixed current** (for example a cable, carrying the current required by the load): $P = R \cdot I^2$. If R increases, **the power increases**: the conductor heats up even more. This is why cables are chosen with a cross-section suited to the current.
- **With a fixed voltage** (for example a resistor connected directly to the mains or to a battery): $P = V^2 / R$. If R increases, **the power decreases** and the heating tends to **stabilise**.

Worked example: the coil cold and hot

The coil from the previous example (10 Ω at 20 °C, 12.34 Ω at 80 °C) runs for 3 h.

Supply	*P* at 20 °C	*P* at 80 °C	Energy in 3 h at 80 °C
fixed current $I = 1$ A	$10 \cdot 1^2 = 10$ W	$12.34 \cdot 1^2 \approx 12.3$ W (**+23 %**)	$12.34 \cdot 3 \approx 37$ Wh
fixed voltage $V = 10$ V	$10^2 / 10 = 10$ W	$10^2 / 12.34 \approx 8.1$ W (**−19 %**)	$8.1 \cdot 3 \approx 24.3$ Wh

When cold, in both cases, 3 h would give $10 \cdot 3 =$ **30 Wh**. With a fixed voltage the current drops too: $I = 10 / 12.34 \approx 0.81$ A.

Energy accounting and cost

To calculate how much is used and how much is spent:

- **1.** read the **power** of each appliance on its rating plate (in W);
- **2.** estimate the **operating hours** per day;
- **3.** calculate the **energy** $E = P \cdot t$ in Wh, then add up the consumptions;
- **4.** convert to **kWh** and multiply by the energy **price**.

> **Cost:** cost [€] = E [kWh] · price [€/kWh]

Worked example: the consumption of a flat

Energy price: **€0.25/kWh**; one month = 30 days.

Appliance	Power	Use per day	Energy per day	Energy per month
LED lamp	10 W	5 h	50 Wh	1.5 kWh
television	100 W	4 h	400 Wh	12 kWh
hair dryer	1200 W	10 min ≈ 0.167 h	200 Wh	6 kWh
computer	150 W	3 h	450 Wh	13.5 kWh
electric oven	2000 W	30 min = 0.5 h	1000 Wh	30 kWh
total			**2100 Wh = 2.1 kWh**	**63 kWh**

- **monthly energy cost:** $63 \cdot 0.25 =$ **€15.75**
- the appliance that weighs most is the **oven**: it has the highest power, even if it stays on only briefly.

The bill also includes fixed charges and taxes: here we calculate only the part due to the energy used.

Worked example: the energy lost in a cable

A 2.5 mm² copper line carries a current of 10 A to a workshop 50 m away. There are two conductors (outgoing and return), so the total length is 100 m and $R =$ **0.7 Ω**, as in the first example. The line works 8 h a day.

- **power lost in the cable:** $P = 0.7 \cdot 10^2 =$ **70 W**

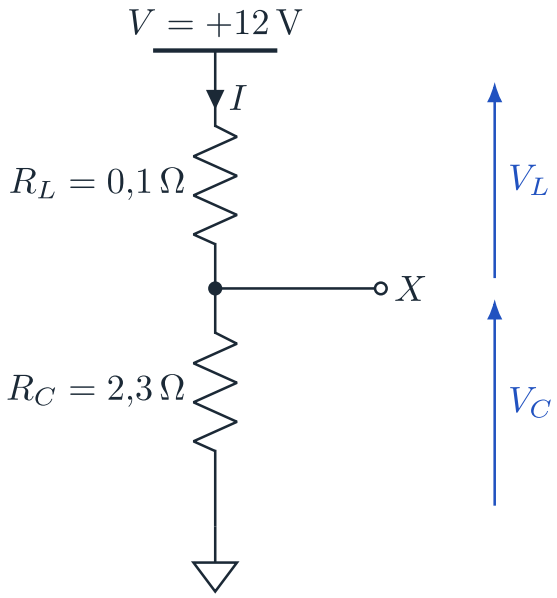
- **energy lost in a day:** 70 · 8 = **560 Wh**
- **energy lost in a month:** 560 · 30 = 16,800 Wh = **16.8 kWh**, that is 16.8 · 0.25 = **€4.20** a month paid just to heat the cable
- **with a 4 mm² copper cable:** R = 0.0175 · 100 / 4 ≈ 0.44 Ω, P ≈ 44 W, losses ≈ 10.5 kWh a month ≈ **€2.63**

A **4 mm² aluminium** cable has the same resistance (0.7 Ω) as 2.5 mm² copper: **same losses**, but it weighs less than half.

The voltage divider: line and load in series

A load supplied through a cable is a **series** circuit: the resistance of the **line** R_L (the two conductors) and the resistance of the **load** R_C. It is exactly a **voltage divider**: the source voltage is shared between the line and the load, and the load receives **less** than the source voltage.

With the divider formula we find the load voltage **straight away**, without calculating the current first.



> **Load voltage:** $V_C = V \cdot R_C / (R_L + R_C)$
>
> **Voltage lost on the line:** $V_L = V \cdot R_L / (R_L + R_C)$
>
> **Efficiency:** $\eta = P_C / P_{tot} = V_C / V$

Worked example: a 12 V system in a camper van

A V = 12 V battery supplies a load with R_C = 2.3 Ω. The cable is 2.5 mm² copper, 7 m outgoing and 7 m return: R_L = 0.0175 · 14 / 2.5 ≈ **0.1 Ω**.

- **load voltage (divider):** V_C = 12 · 2.3 / (0.1 + 2.3) = 12 · 2.3 / 2.4 = **11.5 V**
- **voltage lost on the line:** V_L = 12 · 0.1 / 2.4 = **0.5 V** (check: 11.5 + 0.5 = 12 V ✓)
- **current:** I = 12 / 2.4 = **5 A**

- **powers:** load $P_C = 11.5 \cdot 5 =$ **57.5 W**; line $P_L = 0.1 \cdot 5^2 =$ **2.5 W**; total $12 \cdot 5 = 60$ W ✓
- **efficiency:** $\eta = 57.5 / 60 \approx$ **0.96**, that is 96 %
- **energy in 4 h:** to the load $57.5 \cdot 4 =$ **230 Wh**; lost in the line $2.5 \cdot 4 =$ **10 Wh**; supplied by the battery $60 \cdot 4 =$ **240 Wh**

If the cable is twice as long ($R_L = 0.2\ \Omega$): $V_C = 12 \cdot 2.3 / 2.5 \approx$ **11.04 V**, $I = 12 / 2.5 = 4.8$ A, $P_C \approx 53$ W, $P_L = 0.2 \cdot 4.8^2 \approx 4.6$ W, efficiency **92 %**. The load receives less voltage and the line wastes almost twice as much.

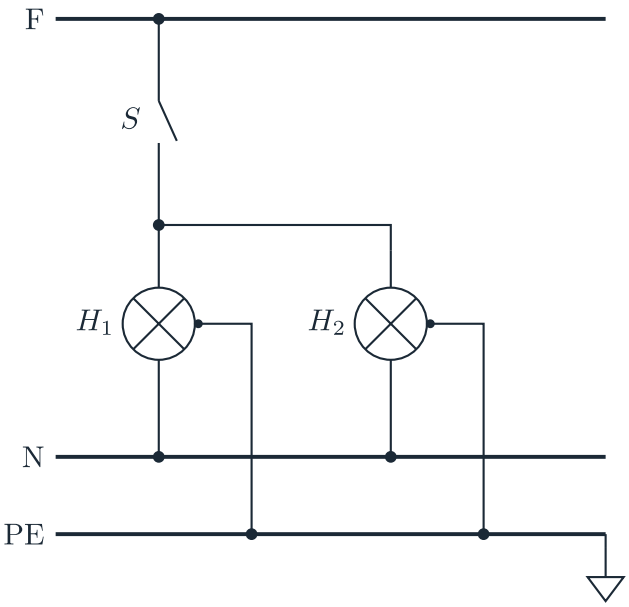
In series **the larger resistance dissipates more**: the current is the same and $P = R \cdot I^2$.

The current divider: loads in parallel

When several loads are connected **in parallel**, they all have the same voltage and the total current is shared among them as in a **current divider**. Each load uses its own share of energy. The most common case is the **electrical installation of a home**.

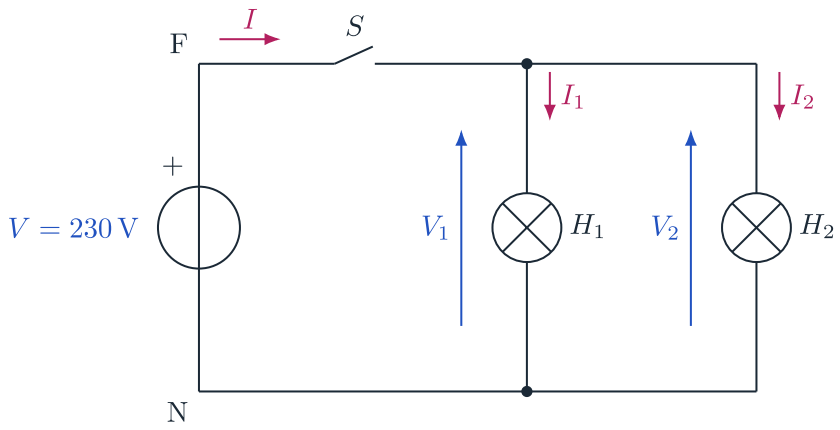
From the installation diagram to the circuit diagram

The **installation diagram** (domestic installation) has three conductors: the **phase** F, the **neutral** N and the **protective conductor** PE, connected to earth. The two lamps H_1 and H_2 are connected between phase and neutral and are switched together by the **switch** S, placed on the phase. The PE is connected to the metal parts of the lamps.



To do the calculations, the same installation is redrawn as a **circuit diagram**:

- the mains, between phase and neutral, becomes a **source** of $V = 230$ V;
- the switch S **closed** is just a piece of wire; **open**, it cuts off the current of both lamps;
- in normal operation the PE **carries no current**, so it does not appear in the circuit diagram;
- the two lamps are connected to the **same two points** (the phase after S and the neutral): they are **in parallel**.



The mains voltage is actually **alternating** (you will study it in the fourth year). For resistive loads, such as incandescent lamps, power and energy are calculated with 230 V as if it were DC.

> **Loads in parallel:** $V_1 = V_2 = V$; $I = I_1 + I_2$
>
> **Current and resistance of a load (rating plate):** $I = P / V$; $R = V^2 / P$

In this circuit, **ideally** (wires with no resistance), the two lamps have the same voltage as the source: $V_1 = V_2 = V = 230$ V.

Worked example: two 75 W lamps

H_1 and H_2 are two **incandescent lamps** with power P = **75 W** at 230 V. They stay on 5 h a day; energy price **€0.25/kWh**, one month = 30 days.

- **current of each lamp:** from $P = V \cdot I$ we get $I = P / V$: $I_1 = 75 / 230 \approx$ **0.33 A**; $I_2 = 75 / 230 \approx$ **0.33 A**
- **total current** (in the phase and in the switch S): $I = I_1 + I_2 = 150 / 230 \approx$ **0.65 A**. Adding the already rounded values gives 0.66 A: the small difference is only due to rounding.
- **resistance of each lamp when on:** $R = V^2 / P = 230^2 / 75 \approx$ **705 Ω**
- **check with the current divider:** $I_1 = I \cdot R_2 / (R_1 + R_2) = 0.65 \cdot 705 / 1410 \approx$ **0.33 A** ✓. With two **equal** loads the current splits **in half**.
- **total power:** $P_{tot} = 75 + 75 =$ **150 W** (check: $V \cdot I = 230 \cdot 0.65 \approx 150$ W ✓)

Energy accounting.

Lamp	Power	Use per day	Energy per day	Energy per month
H_1	75 W	5 h	375 Wh	11.25 kWh
H_2	75 W	5 h	375 Wh	11.25 kWh
total	**150 W**		**750 Wh = 0.75 kWh**	**22.5 kWh**

- **monthly energy cost:** 22.5 · 0.25 ≈ **€5.63**

- **with two 10 W LED lamps**, giving roughly the same light: P_{tot} = 20 W, I = 20 / 230 ≈ 0.09 A, energy 20 · 5 = 100 Wh a day, that is **3 kWh** a month = **€0.75**. The saving is about **€4.88 a month** (87 % less): the incandescent lamp turns almost all its energy into **heat** by the Joule effect, not into light.

Different lamps: where more current flows

Now H_1 is **100 W** and H_2 is **60 W**, still at 230 V.

- **resistances:** $R_1 = 230^2 / 100$ = **529 Ω**; $R_2 = 230^2 / 60$ ≈ **882 Ω**
- **total current:** I = (100 + 60) / 230 ≈ **0.696 A**
- **current divider:** I_1 = 0.696 · 882 / (529 + 882) ≈ **0.435 A**; I_2 = 0.696 · 529 / 1411 ≈ **0.261 A** (check: 100 / 230 ≈ 0.435 A and 60 / 230 ≈ 0.261 A ✓)
- **energy in 5 h:** E_1 = 100 · 5 = **500 Wh**; E_2 = 60 · 5 = **300 Wh**; total **800 Wh**

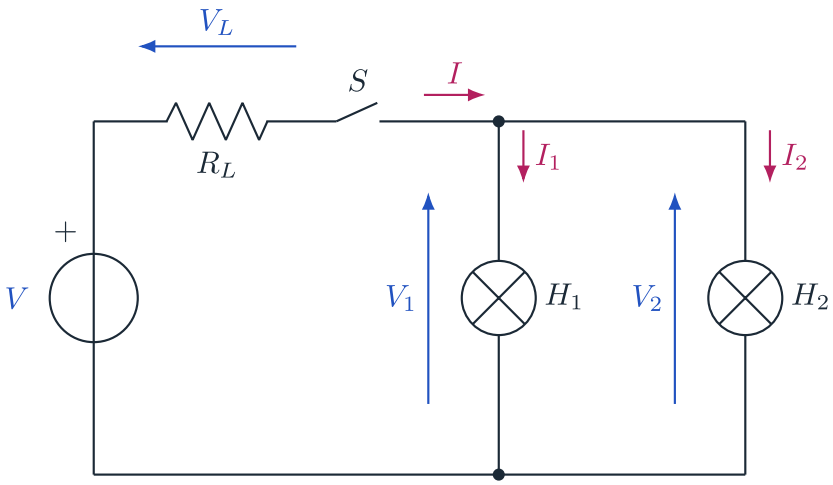
The larger current flows in the lamp with the **smaller resistance**, which is the one with the higher power. In parallel **the smaller resistance dissipates more**: the voltage is the same and $P = V^2 / R$.

Comparison: the same lamps in series. Connected in series at 230 V: I = 230 / (529 + 882) ≈ 0.163 A; $P_1 = 529 \cdot 0.163^2$ ≈ **14 W**; $P_2 = 882 \cdot 0.163^2$ ≈ **23 W**. In series it is the 60 W lamp that dissipates more, and both give very little light (the calculation is only indicative: with less current the filament is cooler and its resistance decreases). This is why loads in a home are connected **in parallel**: each one gets 230 V and can be switched on and off on its own.

In practice: the resistance of the line

The cables carrying the current from the distribution board to the lamps have their own resistance R_L (outgoing and return). The two lamps are still connected to the same two

points, so $V_1 = V_2$, but they are **no longer equal to *V***: part of the voltage is lost on the line. Line and lamps form a **voltage divider**, as in the camper van example.



> **With the line resistance:** $V_1 = V_2 = V \cdot R_P / (R_L + R_P) < V$ (R_P = equivalent resistance of the lamps in parallel)

Example. The two 75 W lamps are 20 m from the distribution board, with a 1.5 mm² copper cable.

- **line resistance:** $L = 2 \cdot 20 = 40$ m; $R_L = 0.0175 \cdot 40 / 1.5 \approx$ **0.47 Ω**
- **resistance of the lamps in parallel:** two equal resistances, $R_P = 705 / 2 \approx$ **353 Ω**
- **voltage across the lamps:** $V_1 = V_2 = 230 \cdot 353 / (0.47 + 353) \approx$ **229.7 V**; about **0.3 V** is lost on the line
- **power lost in the line:** $P_L = 0.47 \cdot 0.65^2 \approx$ **0.2 W**, that is $0.2 \cdot 5 \cdot 30 = 30$ Wh a month

With small loads such as two lamps the loss is negligible: this is why calculations use $V_1 = V_2 = V$. With powerful loads (oven, heater, induction hob) the current is much larger and the voltage drop on the line becomes important: this is why cables are chosen with a cross-section suited to the current.

Formula summary

Resistance of a conductor: $R = \rho \cdot L / S$; round wire $S = \pi \cdot d^2 / 4$; track $S = w \cdot t$

Same resistance with two materials: $S_{Al} / S_{Cu} = \rho_{Al} / \rho_{Cu} = 1.6$

Temperature: $R(T) = R_0 \cdot [1 + \alpha \cdot (T - T_0)]$

Power: $P = V \cdot I$; $P = R \cdot I^2$; $P = V^2 / R$

Heat and energy: $Q = R \cdot I^2 \cdot \Delta t$; $E = P \cdot t$ (W · h = Wh) ; 1 Wh = 3600 J

Cost: cost = E [kWh] · price [€/kWh]

Line and load: $V_C = V \cdot R_C / (R_L + R_C)$; $\eta = V_C / V$

Load (rating plate): $I = P / V$; $R = V^2 / P$

Loads in parallel: $V_1 = V_2 = V$; $I = I_1 + I_2$; $I_1 = I \cdot R_2 / (R_1 + R_2)$; $I_2 = I \cdot R_1 / (R_1 + R_2)$